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1 |----- MODULE Shopping_List -----|
2 EXTENDS Integers, FiniteSets

4 CONSTANTS
5   POSSIBLE_REPLICA_IDS,
6   POSSIBLE_USER_IDS,
7   ASSIGNED_REPLICA,           Users only operate on one replica

9   POSSIBLE_ITEM_IDS,
10  POSSIBLE_ITEM_QUANTITIES,    quantities by which a user can change the quantity of an item
11  POSSIBLE_REQUEST_IDS,       Requests to creator of item to change amount of item

13  NO_ITEM,                     placeholder for item ids that are not yet added to the list
14  NO_REQUEST                  placeholder for request ids that are not yet added to the list

16 VARIABLES
17   replicas, action_counter

19   -----
20   Records -----
21   -----

23 Item  $\triangleq$ 
24   [id : POSSIBLE_ITEM_IDS,
25    creator : POSSIBLE_USER_IDS,
26    quantity : Nat,           Quantity changes via request, only creator can accept/deny
27    version : Nat,           Used only to track quantity of item
28    deleted_counter : Nat ]   deleted is causal length counter

30 Request  $\triangleq$ 
31   [id : POSSIBLE_REQUEST_IDS,
32    sender : POSSIBLE_USER_IDS,
33    change_amount : Int,
34    processed : BOOLEAN ]     Used to ensure a request is only processed once (  $\rightarrow$  double counting)

36 Replica  $\triangleq$ 
37   [id : POSSIBLE_REPLICA_IDS,
38    recorded_items : [POSSIBLE_ITEM_IDS  $\rightarrow$  (Item  $\cup$  {NO_ITEM})],
39    recorded_requests : [POSSIBLE_REQUEST_IDS  $\rightarrow$  (Request  $\cup$  {NO_REQUEST})] ]

42   -----
43   Initialization -----
44   -----

46 Init  $\triangleq$ 
47    $\wedge$  replicas =
48     [rid  $\in$  POSSIBLE_REPLICA_IDS  $\mapsto$ 
49       [id  $\mapsto$  rid,
50        recorded_items  $\mapsto$  [iid  $\in$  POSSIBLE_ITEM_IDS  $\mapsto$  NO_ITEM],
51        recorded_requests  $\mapsto$  [rqid  $\in$  POSSIBLE_REQUEST_IDS  $\mapsto$  NO_REQUEST]
52      ]
53   ]
54    $\wedge$  action_counter = 0

57   -----
58   Helper Functions -----
59   -----

61   Causal length logic
62   Is_deleted(item)  $\triangleq$ 
63     item.deleted_counter%2 = 1

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66
67 Item actions
68
70 Add_item  $\triangleq$ 
71    $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
72    $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
73    $\exists actor \in POSSIBLE\_USER\_IDs :$ 
74    $\exists item\_quantity \in POSSIBLE\_ITEM\_QUANTITIES :$ 
75    $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
76   No other replica has item with same id (uniqueness, in real implementation use dynamic ids)
77    $\wedge \forall other\_rid \in POSSIBLE\_REPLICA\_IDs : replicas[other\_rid].recorded\_items[iid] = NO\_ITEM$ 
79    $\wedge LET new\_item \triangleq$ 
80      $[id \mapsto iid,$ 
81        $creator \mapsto actor,$ 
82        $quantity \mapsto item\_quantity,$ 
83        $version \mapsto 1,$ 
84        $deleted\_counter \mapsto 0$ 
85      $]$ 
86    $new\_replica \triangleq$ 
87    $[replicas[rid] \text{ EXCEPT } !.recorded\_items = [@ \text{ EXCEPT } ![iid] = new\_item]]$ 
88   IN
89    $\wedge replicas' = [replicas \text{ EXCEPT } ![rid] = new\_replica]$ 
90    $\wedge action\_counter' = action\_counter + 1$ 
92
93 Request_quantity_increase  $\triangleq$ 
94    $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
95    $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
96    $\exists rrid \in POSSIBLE\_REQUEST\_IDs :$ 
97    $\exists actor \in POSSIBLE\_USER\_IDs :$ 
98    $\exists quantity \in POSSIBLE\_ITEM\_QUANTITIES :$ 
99    $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
100    $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
101   No other replica has request with same id (uniqueness, in real implementation use dynamic ids)
102    $\wedge \forall other\_rid \in POSSIBLE\_REPLICA\_IDs : replicas[other\_rid].recorded\_requests[rrid] = NO\_REQUEST$ 
104    $\wedge LET new\_request \triangleq$ 
105      $[id \mapsto rrid,$ 
106        $sender \mapsto actor,$ 
107        $change\_amount \mapsto quantity,$ 
108        $processed \mapsto FALSE$ 
109      $]$ 
110    $new\_replica \triangleq$ 
111    $[replicas[rid] \text{ EXCEPT } !.recorded\_requests = [@ \text{ EXCEPT } ![rrid] = new\_request]]$ 
112   IN
113    $\wedge replicas' = [replicas \text{ EXCEPT } ![rid] = new\_replica]$ 
114    $\wedge action\_counter' = action\_counter + 1$ 
116
117 Request_quantity_decrease  $\triangleq$ 
118    $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
119    $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
120    $\exists rrid \in POSSIBLE\_REQUEST\_IDs :$ 
121    $\exists actor \in POSSIBLE\_USER\_IDs :$ 
122    $\exists quantity \in POSSIBLE\_ITEM\_QUANTITIES :$ 
123    $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
124    $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
125   No other replica has request with same id (uniqueness, in real implementation use dynamic ids)

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126       $\wedge \forall other\_rid \in POSSIBLE\_REPLICA\_IDs : replicas[other\_rid].recorded\_requests[rrid] = NO\_REQUEST$ 
128       $\wedge LET\ new\_request \triangleq$ 
129           $[id \mapsto rrid,$ 
130              $sender \mapsto actor,$ 
131              $change\_amount \mapsto quantity * (-1),$ 
132              $processed \mapsto FALSE$ 
133           $]$ 
134       $new\_replica \triangleq$ 
135           $[replicas[rid] \text{ EXCEPT } !.recorded\_requests = [@ \text{ EXCEPT } ![rrid] = new\_request]]$ 
136      IN
137           $\wedge replicas' = [replicas \text{ EXCEPT } ![rid] = new\_replica]$ 
138           $\wedge action\_counter' = action\_counter + 1$ 

141      Accept quantity change request if quantity won't go negative
142       $Accept\_request \triangleq$ 
143           $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
144           $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
145           $\exists rrid \in POSSIBLE\_REQUEST\_IDs :$ 
146           $\exists actor \in POSSIBLE\_USER\_IDs :$ 
147               $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
148               $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
149               $\wedge replicas[rid].recorded\_items[iid].creator = actor$ 
150               $\wedge replicas[rid].recorded\_requests[rrid] \neq NO\_REQUEST$ 
151               $\wedge replicas[rid].recorded\_requests[rrid].processed = FALSE$ 

153           $\wedge (replicas[rid].recorded\_items[iid].quantity + replicas[rid].recorded\_requests[rrid].change\_amount) \geq 0$ 
154           $\wedge LET$ 
155               $current\_item \triangleq replicas[rid].recorded\_items[iid]$ 
156               $current\_req \triangleq replicas[rid].recorded\_requests[rrid]$ 

158               $new\_item \triangleq$ 
159                   $[current\_item \text{ EXCEPT } !.quantity = @ + current\_req.change\_amount,$ 
160                      $!.version = @ + 1]$ 
161               $new\_request \triangleq$ 
162                   $[current\_req \text{ EXCEPT } !.processed = TRUE]$ 
163               $new\_replica \triangleq$ 
164                   $[replicas[rid] \text{ EXCEPT } !.recorded\_items[iid] = new\_item,$ 
165                      $!.recorded\_requests[rrid] = new\_request]$ 
166          IN
167               $\wedge replicas' = [replicas \text{ EXCEPT } ![rid] = new\_replica]$ 
168               $\wedge action\_counter' = action\_counter + 1$ 

171      Deny quantity change request (for any reason)
172       $Deny\_request \triangleq$ 
173           $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
174           $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
175           $\exists rrid \in POSSIBLE\_REQUEST\_IDs :$ 
176           $\exists actor \in POSSIBLE\_USER\_IDs :$ 
177               $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
178               $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
179               $\wedge replicas[rid].recorded\_items[iid].creator = actor$ 
180               $\wedge replicas[rid].recorded\_requests[rrid] \neq NO\_REQUEST$ 
181               $\wedge replicas[rid].recorded\_requests[rrid].processed = FALSE$ 

183           $\wedge LET$ 
184               $current\_req \triangleq replicas[rid].recorded\_requests[rrid]$ 
186               $new\_request \triangleq$ 

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187      [current_req EXCEPT !.processed = TRUE]
188      new_replica  $\triangleq$ 
189      [replicas[rid] EXCEPT !.recorded_requests[rrid] = new_request]
190  IN
191       $\wedge$  replicas' = [replicas EXCEPT ![rid] = new_replica]
192       $\wedge$  action_counter' = action_counter + 1

195  Delete_item  $\triangleq$ 
196       $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
197       $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
198       $\exists actor \in POSSIBLE\_USER\_IDs :$ 
199       $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
200       $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
201       $\wedge \neg Is\_deleted(replicas[rid].recorded\_items[iid])$ 
202
203       $\wedge replicas' = [replicas EXCEPT ![rid].recorded\_items[iid].deleted\_counter = @ + 1]$ 
204       $\wedge action\_counter' = action\_counter + 1$ 

207  Reinstate_item  $\triangleq$ 
208       $\exists iid \in POSSIBLE\_ITEM\_IDs :$ 
209       $\exists rid \in POSSIBLE\_REPLICA\_IDs :$ 
210       $\exists actor \in POSSIBLE\_USER\_IDs :$ 
211       $\wedge ASSIGNED\_REPLICA[actor] = rid$ 
212       $\wedge replicas[rid].recorded\_items[iid] \neq NO\_ITEM$ 
213       $\wedge Is\_deleted(replicas[rid].recorded\_items[iid])$ 
214
215       $\wedge replicas' = [replicas EXCEPT ![rid].recorded\_items[iid].deleted\_counter = @ + 1]$ 
216       $\wedge action\_counter' = action\_counter + 1$ 

219  
220  Merge helper 
221  

223  Merge_item(item_own, item_other)  $\triangleq$ 
224      

If only one replica has the item with id keep this one


225      IF item_own = NO_ITEM  $\wedge$  item_other = NO_ITEM
226      THEN NO_ITEM
227      ELSE IF item_own  $\neq$  NO_ITEM  $\wedge$  item_other = NO_ITEM
228      THEN item_own
229      ELSE IF item_own = NO_ITEM  $\wedge$  item_other  $\neq$  NO_ITEM
230      THEN item_other
231      

If both replicas have item with id apply merge logic


232      ELSE
233      LET
234          

Determine which record holds the most recent version for quantity


235          higher_version_item  $\triangleq$  IF item_own.version  $\geq$  item_other.version
236          THEN item_own
237          ELSE item_other
238
239          

Delete is concurrently updated independent of the version logic (use cls merge)


240          merged_deleted  $\triangleq$  CHOOSE  $n \in \{item\_own.deleted\_counter, item\_other.deleted\_counter\} :$ 
241           $n \geq item\_own.deleted\_counter \wedge n \geq item\_other.deleted\_counter$ 
242      IN
243          [item_own EXCEPT !.quantity = higher_version_item.quantity,
244           !.version = higher_version_item.version,
245           !.deleted_counter = merged_deleted]

248  Merge_request(req_own, req_other)  $\triangleq$ 

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249 IF  $req\_own = NO\_REQUEST \wedge req\_other = NO\_REQUEST$ 
250 THEN  $NO\_REQUEST$ 
251 ELSE IF  $req\_own \neq NO\_REQUEST \wedge req\_other = NO\_REQUEST$ 
252 THEN  $req\_own$ 
253 ELSE IF  $req\_own = NO\_ITEM \wedge req\_other \neq NO\_REQUEST$ 
254 THEN  $req\_other$ 
255 ELSE IF  $req\_own = NO\_REQUEST \wedge req\_other \neq NO\_REQUEST$ 
256 THEN  $req\_other$ 
257 ELSE Prefer true in merges
258  $[req\_own \text{ EXCEPT } !.processed = req\_own.processed \vee req\_other.processed]$ 

260
261 Merge action
262

264  $Merge\_replicas \triangleq$ 
265  $\exists own\_rid, other\_rid \in POSSIBLE\_REPLICA\_IDs :$ 
266  $\wedge own\_rid \neq other\_rid$ 
267  $\wedge \text{LET}$ 
268  $own\_items \triangleq replicas[own\_rid].recorded\_items$ 
269  $other\_items \triangleq replicas[other\_rid].recorded\_items$ 
270  $own\_requests \triangleq replicas[own\_rid].recorded\_requests$ 
271  $other\_requests \triangleq replicas[other\_rid].recorded\_requests$ 
272
273  $merged\_recorded\_items \triangleq [iid \in POSSIBLE\_ITEM\_IDs \mapsto$ 
274  $Merge\_item(own\_items[iid], other\_items[iid])]$ 
275
276  $merged\_recorded\_requests \triangleq [rrid \in POSSIBLE\_REQUEST\_IDs \mapsto$ 
277  $Merge\_request(own\_requests[rrid], other\_requests[rrid])]$ 
278 IN
279  $\wedge replicas' = [replicas \text{ EXCEPT } ![own\_rid].recorded\_items = merged\_recorded\_items,$ 
280  $! [own\_rid].recorded\_requests = merged\_recorded\_requests]$ 
281  $\wedge \text{UNCHANGED } action\_counter$ 

284
285 Next action
286

288  $Next \triangleq$ 
289  $\vee Add\_item$ 
290  $\vee Request\_quantity\_increase$ 
291  $\vee Request\_quantity\_decrease$ 
292  $\vee Deny\_request$ 
293  $\vee Accept\_request$ 
294  $\vee Delete\_item$ 
295  $\vee Reinstate\_item$ 
296  $\vee Merge\_replicas$ 
297  $\vee \text{UNCHANGED } \langle replicas, action\_counter \rangle$ 

300
301 Invariants
302

304  $TypeOK \triangleq$ 
305  $\forall rid \in POSSIBLE\_REPLICA\_IDs :$ 
306  $\wedge replicas[rid].recorded\_items \in [POSSIBLE\_ITEM\_IDs \rightarrow (Item \cup \{NO\_ITEM\})]$ 
307  $\wedge replicas[rid].recorded\_requests \in [POSSIBLE\_REQUEST\_IDs \rightarrow (Request \cup \{NO\_REQUEST\})]$ 

310  $Quantities\_non\_negative \triangleq$ 

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311 $\forall rid \in POSSIBLE_REPLICA_IDs :$
 312 $\forall iid \in POSSIBLE_ITEM_IDs :$
 313 $replicas[rid].recorded_items[iid] \neq NO_ITEM \Rightarrow$
 314 $replicas[rid].recorded_items[iid].quantity \geq 0$

317

 318 Safety Helpers

 319

321 The causal length counter for deletion/reinstate only increases
 322 $No_decrease_deleted_counter \triangleq$
 323 $\forall rid \in POSSIBLE_REPLICA_IDs, iid \in POSSIBLE_ITEM_IDs :$
 324 $(replicas[rid].recorded_items[iid] \neq NO_ITEM)$
 325 $\Rightarrow$
 326 $\wedge replicas'[rid].recorded_items[iid] \neq NO_ITEM$
 327 $\wedge replicas'[rid].recorded_items[iid].deleted_counter$
 328 $\geq replicas[rid].recorded_items[iid].deleted_counter$

330 The version counter for items only increases
 331 $No_decrease_version_counter \triangleq$
 332 $\forall rid \in POSSIBLE_REPLICA_IDs, iid \in POSSIBLE_ITEM_IDs :$
 333 $(replicas[rid].recorded_items[iid] \neq NO_ITEM)$
 334 $\Rightarrow$
 335 $\wedge replicas'[rid].recorded_items[iid] \neq NO_ITEM$
 336 $\wedge replicas'[rid].recorded_items[iid].version$
 337 $\geq replicas[rid].recorded_items[iid].version$

339 Processed requests stay processed
 340 $Processed_requests_terminal \triangleq$
 341 $\forall rid \in POSSIBLE_REPLICA_IDs, rrid \in POSSIBLE_REQUEST_IDs :$
 342 $(replicas[rid].recorded_requests[rrid] \neq NO_REQUEST \wedge replicas[rid].recorded_requests[rrid].processed = \text{TRUE})$
 343 $\Rightarrow$
 344 $\wedge replicas'[rid].recorded_requests[rrid] \neq NO_REQUEST$
 345 $\wedge replicas'[rid].recorded_requests[rrid].processed = \text{TRUE}$

348

 349 Additional Safety

 350

352 Monotinic growth of causal length counter
 353 $Safety_deleted_counter_non_decreasing \triangleq$
 354 $\Box[No_decrease_deleted_counter]_{\{\langle replicas, action_counter \rangle\}}$

356 Monotinic growth of version counter
 357 $Safety_version_counter_non_decreasing \triangleq$
 358 $\Box[No_decrease_version_counter]_{\{\langle replicas, action_counter \rangle\}}$

360 Processed requests terminal
 361 $Safety_processed_requests_terminal \triangleq$
 362 $\Box[Processed_requests_terminal]_{\{\langle replicas, action_counter \rangle\}}$

366

 367 Liveness helpers

 368

370 $All_replicas_have_at_least_item_version(iid, version) \triangleq$
 371 $\forall rid \in POSSIBLE_REPLICA_IDs :$
 372 $\wedge replicas[rid].recorded_items[iid] \neq NO_ITEM$
 373 $\wedge replicas[rid].recorded_items[iid].version \geq version$

375 $All_replicas_have_at_least_deleted_counter(iid, counter) \triangleq$
 376 $\forall rid \in POSSIBLE_REPLICA_IDs :$
 377 $\wedge replicas[rid].recorded_items[iid] \neq NO_ITEM$
 378 $\wedge replicas[rid].recorded_items[iid].deleted_counter \geq counter$

 380 $All_replicas_have_item(iid) \triangleq$
 381 $\forall rid \in POSSIBLE_REPLICA_IDs :$
 382 $replicas[rid].recorded_items[iid] \neq NO_ITEM$

 385 $\overline{\overline{\hspace{10em}}}$
 386 Liveness $\overline{\overline{\hspace{10em}}}$
 387 $\overline{\overline{\hspace{10em}}}$

 389 1. Always eventually all replicas need to catch up to the latest version of an item,
 390 so most recent quantity updates reach every replica
 391 $Liveness_versions_propagate \triangleq$
 392 $\forall rid \in POSSIBLE_REPLICA_IDs, iid \in POSSIBLE_ITEM_IDs :$
 393 $\Box \Diamond (replicas[rid].recorded_items[iid] \neq NO_ITEM$
 394 $\Rightarrow All_replicas_have_at_least_item_version(iid, replicas[rid].recorded_items[iid].version))$

 396 2. Always eventually all replicas need to agree if an item is present/removed;
 397 This is independent from the version as we allow concurrent add/remove
 398 $Liveness_Deletion_state_propagates \triangleq$
 399 $\forall rid \in POSSIBLE_REPLICA_IDs, iid \in POSSIBLE_ITEM_IDs :$
 400 $\Box \Diamond (replicas[rid].recorded_items[iid] \neq NO_ITEM$
 401 $\Rightarrow All_replicas_have_at_least_deleted_counter(iid, replicas[rid].recorded_items[iid].deleted_counter))$

 403 3. Every added item is eventually known by all replicas and remains known thereafter
 404 $Liveness_Item_presence_propagates \triangleq$
 405 $\forall rid \in POSSIBLE_REPLICA_IDs, iid \in POSSIBLE_ITEM_IDs :$
 406 $(replicas[rid].recorded_items[iid] \neq NO_ITEM)$
 407 $\Rightarrow \Diamond \Box All_replicas_have_item(iid)$

 410 $\overline{\overline{\hspace{10em}}}$
 411 Specification $\overline{\overline{\hspace{10em}}}$
 412 $\overline{\overline{\hspace{10em}}}$
 413 $Spec \triangleq Init \wedge \Box [Next]_{\langle replicas, action_counter \rangle}$

 415 $FairSpec \triangleq$
 416 $\wedge Spec$
 417 $\wedge WF_{\langle replicas, action_counter \rangle}(Merge_replicas)$

420 $\overline{\hspace{10em}}$
 \ * Modification History
 \ * Last modified Mon May 25 08:57:52 CEST 2026 by floyd
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